Akilan D

Step 1: Assing ip for all Routers and add static route for all Routers

Router 1:

```
R1#show ip interface brief
Interface                  IP-Address      OK? Method Status                Prot
ocol
Ethernet0/0                10.0.0.1        YES manual up                    up

Ethernet0/1                20.0.0.1        YES manual up                    up

Ethernet0/2                unassigned      YES TFTP   down                  down

Ethernet0/3                unassigned      YES TFTP   down                  down

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        10.0.0.0/8 is directly connected, Ethernet0/0
L        10.0.0.1/32 is directly connected, Ethernet0/0
      15.0.0.0/24 is subnetted, 1 subnets
S        15.0.0.0 [1/0] via 20.0.0.2
      20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        20.0.0.0/8 is directly connected, Ethernet0/1
L        20.0.0.1/32 is directly connected, Ethernet0/1
      21.0.0.0/24 is subnetted, 1 subnets
S        21.0.0.0 [1/0] via 20.0.0.2
      30.0.0.0/24 is subnetted, 1 subnets
S        30.0.0.0 [1/0] via 20.0.0.2
      162.14.0.0/24 is subnetted, 1 subnets
S        162.14.0.0 [1/0] via 20.0.0.2
      172.19.0.0/24 is subnetted, 1 subnets
S        172.19.0.0 [1/0] via 20.0.0.2
S     192.168.168.0/24 [1/0] via 20.0.0.2
R1#
```

Router 2:

```
R2#sh ip int bri
Interface                  IP-Address      OK? Method Status                Prot
ocol
Ethernet0/0                20.0.0.2        YES manual up                    up

Ethernet0/1                30.0.0.1        YES manual up                    up

Ethernet0/2                unassigned      YES TFTP   down                  down

Ethernet0/3                unassigned      YES TFTP   down                  down

R2#sh ip ro
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      10.0.0.0/24 is subnetted, 1 subnets
S        10.0.0.0 [1/0] via 20.0.0.1
      15.0.0.0/24 is subnetted, 1 subnets
S        15.0.0.0 [1/0] via 30.0.0.2
      20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        20.0.0.0/24 is directly connected, Ethernet0/0
L        20.0.0.2/32 is directly connected, Ethernet0/0
      21.0.0.0/24 is subnetted, 1 subnets
S        21.0.0.0 [1/0] via 30.0.0.2
      30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        30.0.0.0/24 is directly connected, Ethernet0/1
L        30.0.0.1/32 is directly connected, Ethernet0/1
      162.14.0.0/24 is subnetted, 1 subnets
S        162.14.0.0 [1/0] via 30.0.0.2
      172.19.0.0/24 is subnetted, 1 subnets
S        172.19.0.0 [1/0] via 30.0.0.2
S     192.168.168.0/24 [1/0] via 30.0.0.2
```

Router 3:

```
R3#sh ip int bri
Interface                  IP-Address      OK? Method Status                Prot
ocol
Ethernet0/0                30.0.0.2        YES manual up                    up

Ethernet0/1                172.19.0.1      YES manual up                    up

Ethernet0/2                21.0.0.1        YES manual up                    up

Ethernet0/3                unassigned      YES TFTP   down                  down

R3#sh ip ro
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      10.0.0.0/24 is subnetted, 1 subnets
S        10.0.0.0 [1/0] via 30.0.0.1
      15.0.0.0/24 is subnetted, 1 subnets
S        15.0.0.0 [1/0] via 172.19.0.2
      20.0.0.0/24 is subnetted, 1 subnets
S        20.0.0.0 [1/0] via 30.0.0.1
      21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        21.0.0.0/24 is directly connected, Ethernet0/2
L        21.0.0.1/32 is directly connected, Ethernet0/2
      30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        30.0.0.0/24 is directly connected, Ethernet0/0
L        30.0.0.2/32 is directly connected, Ethernet0/0
      162.14.0.0/24 is subnetted, 1 subnets
S        162.14.0.0 [1/0] via 21.0.0.2
      172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        172.19.0.0/24 is directly connected, Ethernet0/1
L        172.19.0.1/32 is directly connected, Ethernet0/1
S     192.168.168.0/24 [1/0] via 21.0.0.2
R3#
```

Router 4:

```
R4>sh ip int bri
Interface                  IP-Address      OK? Method Status                Prot
ocol
Ethernet0/0                15.0.0.1        YES manual up                    up

Ethernet0/1                172.19.0.2      YES manual up                    up

Ethernet0/2                unassigned      YES unset  administratively down down

Ethernet0/3                unassigned      YES unset  administratively down down

R4>sh ip ro
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      10.0.0.0/24 is subnetted, 1 subnets
S        10.0.0.0 [1/0] via 172.19.0.1
      15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        15.0.0.0/24 is directly connected, Ethernet0/0
L        15.0.0.1/32 is directly connected, Ethernet0/0
      20.0.0.0/24 is subnetted, 1 subnets
S        20.0.0.0 [1/0] via 172.19.0.1
      21.0.0.0/24 is subnetted, 1 subnets
S        21.0.0.0 [1/0] via 172.19.0.1
      30.0.0.0/24 is subnetted, 1 subnets
S        30.0.0.0 [1/0] via 172.19.0.1
      162.14.0.0/24 is subnetted, 1 subnets
S        162.14.0.0 [1/0] via 172.19.0.1
      172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        172.19.0.0/24 is directly connected, Ethernet0/1
L        172.19.0.2/32 is directly connected, Ethernet0/1
S     192.168.168.0/24 [1/0] via 172.19.0.1
```

Router 5:

```
R5#sh ip int bri
Interface                  IP-Address      OK? Method Status                Prot
ocol
Ethernet0/0                21.0.0.2        YES manual up                    up

Ethernet0/1                192.168.168.1   YES manual up                    up

Ethernet0/2                unassigned      YES TFTP   down                  down

Ethernet0/3                unassigned      YES TFTP   down                  down

R5#sh ip ro
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      10.0.0.0/24 is subnetted, 1 subnets
S        10.0.0.0 [1/0] via 21.0.0.1
      15.0.0.0/24 is subnetted, 1 subnets
S        15.0.0.0 [1/0] via 21.0.0.1
      20.0.0.0/24 is subnetted, 1 subnets
S        20.0.0.0 [1/0] via 21.0.0.1
      21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        21.0.0.0/24 is directly connected, Ethernet0/0
L        21.0.0.2/32 is directly connected, Ethernet0/0
      30.0.0.0/24 is subnetted, 1 subnets
S        30.0.0.0 [1/0] via 21.0.0.1
      162.14.0.0/24 is subnetted, 1 subnets
S        162.14.0.0 [1/0] via 192.168.168.2
      172.19.0.0/24 is subnetted, 1 subnets
S        172.19.0.0 [1/0] via 21.0.0.1
      192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.168.0/24 is directly connected, Ethernet0/1
L        192.168.168.1/32 is directly connected, Ethernet0/1
```

Router 6:

```
R6#sh ip int bri
Interface                  IP-Address      OK? Method Status                Prot
ocol
Ethernet0/0                162.14.0.1      YES manual up                    up

Ethernet0/1                192.168.168.2   YES manual up                    up

Ethernet0/2                unassigned      YES TFTP   down                  down

Ethernet0/3                unassigned      YES TFTP   down                  down

R6#sh ip ro
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      10.0.0.0/24 is subnetted, 1 subnets
S        10.0.0.0 [1/0] via 192.168.168.1
      15.0.0.0/24 is subnetted, 1 subnets
S        15.0.0.0 [1/0] via 192.168.168.1
      20.0.0.0/24 is subnetted, 1 subnets
S        20.0.0.0 [1/0] via 192.168.168.1
      21.0.0.0/24 is subnetted, 1 subnets
S        21.0.0.0 [1/0] via 192.168.168.1
      30.0.0.0/24 is subnetted, 1 subnets
S        30.0.0.0 [1/0] via 192.168.168.1
      162.14.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        162.14.0.0/24 is directly connected, Ethernet0/0
L        162.14.0.1/32 is directly connected, Ethernet0/0
      172.19.0.0/24 is subnetted, 1 subnets
S        172.19.0.0 [1/0] via 192.168.168.1
      192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.168.0/24 is directly connected, Ethernet0/1
L        192.168.168.2/32 is directly connected, Ethernet0/1
```

Step 2: Assign ip for all Three pc and ping another Network Pc's

```
C:\Documents and Settings\user>ncpa.cpl

C:\Documents and Settings\user>ipconfig

Windows IP Configuration


Ethernet adapter Local Area Connection:

        Connection-specific DNS Suffix  . :
        IP Address. . . . . . . . . . . . : 10.0.0.10
        Subnet Mask . . . . . . . . . . . : 255.0.0.0
        Default Gateway . . . . . . . . . : 10.0.0.1

C:\Documents and Settings\user>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time=4ms TTL=124
Reply from 15.0.0.10: bytes=32 time=1ms TTL=124
Reply from 15.0.0.10: bytes=32 time=1ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 15.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 4ms, Average = 2ms

C:\Documents and Settings\user>ping 162.14.0.10

Pinging 162.14.0.10 with 32 bytes of data:

Reply from 162.14.0.10: bytes=32 time=3ms TTL=123
Reply from 162.14.0.10: bytes=32 time=3ms TTL=123
Reply from 162.14.0.10: bytes=32 time=2ms TTL=123
Reply from 162.14.0.10: bytes=32 time=2ms TTL=123

Ping statistics for 162.14.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>_
```

```
C:\Documents and Settings\user>ipconfig

Windows IP Configuration


Ethernet adapter Local Area Connection:

        Connection-specific DNS Suffix  . :
        IP Address. . . . . . . . . . . . : 15.0.0.10
        Subnet Mask . . . . . . . . . . . : 255.0.0.0
        Default Gateway . . . . . . . . . : 15.0.0.1

C:\Documents and Settings\user>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=1ms TTL=124
Reply from 10.0.0.10: bytes=32 time=3ms TTL=124
Reply from 10.0.0.10: bytes=32 time=1ms TTL=124
Reply from 10.0.0.10: bytes=32 time=1ms TTL=124

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 3ms, Average = 1ms

C:\Documents and Settings\user>ping 162.14.0.10

Pinging 162.14.0.10 with 32 bytes of data:

Reply from 162.14.0.10: bytes=32 time=1ms TTL=124
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124
Reply from 162.14.0.10: bytes=32 time=1ms TTL=124
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 162.14.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\user>
```

```
C:\Documents and Settings\user>ipconfig

Windows IP Configuration


Ethernet adapter Local Area Connection:

        Connection-specific DNS Suffix  . :
        IP Address. . . . . . . . . . . . : 162.14.0.10
        Subnet Mask . . . . . . . . . . . : 255.255.0.0
        Default Gateway . . . . . . . . . : 162.14.0.1

C:\Documents and Settings\user>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=2ms TTL=123
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123
Reply from 10.0.0.10: bytes=32 time=1ms TTL=123
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\user>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time=1ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 15.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\user>
```